

Advanced Power Electronic Conversion Technologies

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Abstract— Generation, storage, and distribution of electric energy all depend on power electronics, whose robustness and efficiency have historically been limited by conventional Silicon (Si) devices. A revolution is anticipated with the introducing of newest technologies in today's life. These developments fuel the creation of highly specialized conversion topologies as well as multi-level architectures. Additionally, advanced modulation and control methods, particularly Selective Harmonic Mitigation and Model Predictive Control, are being used to maximize performance and adhere to increasingly stringent grid codes, are always changing to take advantage of topological redundancies, control device limitations, and enhance output quality. The state-of-the-art and future directions in these specialized power electronics fields are covered in this report.

Keywords- Power Electronics, converter topologies, Electric motor vehicle, Renewable system energy, predictive system control.

I. INTRODUCTION

Considering that electric energy accounts for 40% of global energy consumption, power electronics are essential to the energy cycle. Losses dissipated in power semiconductor devices are the main factor limiting the efficiency of power electronic converters [1]. Existing Si-based power devices have significant drawbacks, such as a maximum commercial IGBT breakdown voltage of 6.5 kV and limited operation below 200 °C. These physical constraints lower converter efficiency, frequently requiring passive parts and intricate, costly cooling systems [2].

These limitations should be overcome by a new technology, which would result in use of electric energy more sensibly and a significant increase in the size and robustness of power converters [3].

Continuous improvements in conversion topologies are necessary to meet the demand for improved performance across a variety of industries, including transportation (EVs, HEVs),

and renewable energy generation (photovoltaic (PV) systems, wind turbines) [4][5].

In order to achieve arbitrary output fundamental components while addressing issues like harmonic distortion and device limitations, modulation techniques have evolved in tandem with the growing complexity of Voltage-Source Inverter (VSI) architectures [6][7].

II. WIDE BANDGAP DEVICES FOR POWER CONVERTERS

For a high frequency (HF), high voltage (HV) electronics, WBG semiconductors—in particular, SiC and GaN—are the subject of intense research and development. The need to boost efficiency and convert power more sensibly is what is driving the switch to WBG materials [1],[2].

A. Silicon Carbide (SiC) Devices

The commercial availability and process maturity of SiC make it highly suitable for power applications [5].

1. SiC Rectifiers and Diodes:

Since going on sale in 2001, SiC Schottky Barrier Diodes (SBDs) have steadily increased in rating, reaching 1.2 kV and 1.7 kV with current capacity up to 50A. Si p-i-n rectifiers in the 600V-3kV range are anticipated to be replaced by commercial 3.3kV Schottky diodes, which have been announced. SiC SBDs are perfect as free-wheeling diodes with Si IGBTs because they have a lower reverse-recovery charge than ultrafast Si p-i-n diodes. Additionally, SiC devices are candidates for use in space and aerospace missions due to their capacity for high-temperature operation [5],[6].

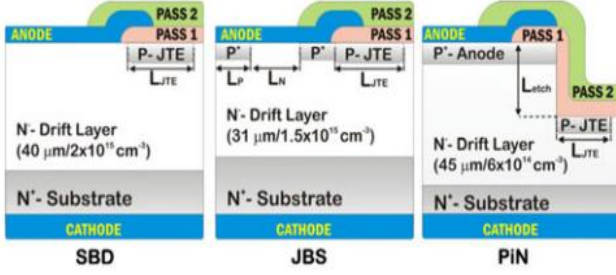


Fig. 1 Cross section of different diodes [4].

2. SiC Switching Devices (JFETs, MOSFETs, IGBTs):

- JFETs: Due to its capability to function at high temperatures and also frequencies, the SiC Junction Field-Effect Transistor (JFET) is an alternative for the kV voltage range. Commercially available hybrid switches achieve normally-off behavior by combining a Si MOSFET with a vertical SiC normally-on JFET in a cascode configuration. However, these hybrid switches' capability to work at high temperatures is restricted by the low-voltage Si MOSFET [6].
- MOSFETs: There are commercially available 4H-SiC Double-Implanted MOSFETs with demonstrations up to 10kV in the 600V–1.2kV range. For operation above 100kHz, companies are releasing many newer technologies that can combine multiple technologies along with SiC MOSFETs [5],[6].
- IGBTs: There have been reports of ultra high voltage 12kV 4 H-SiC IGBTs. At a 2kV dc bus with a 120kW output power, the application of SiC Gate Turn-off Thyristors in back-to-back inverter systems has been demonstrated. [6],[7].

B. Gallium Nitride (GaN) Devices

GaN's material characteristics make it is preferable because it provides a less expensive technology and enables material growth on substrates with large diameters. [7].

1. HEMTs and Performance

For high mobility use a 2-D Electron Gas. These devices offer a potential 100-fold performance advantage over silicon power devices based on the figure of merit. Blocking voltages of more than 1 kV have been attained by high-performance GaN/GaN HEMTs on Si-substrates, whereas devices on sapphire substrates have reached 8.3kV [5],[7].

2. Normally-Off Operation:

GaN HEMTs are inherently on depletion mode. Since normally-off switches are preferred in power systems, research focuses on:

- Using a recessed-gate structure to induce a positively of threshold voltage.
- Changing the threshold voltage in the gate region by applying fluorine-based plasma treatment.
- A p-n junction gate's selective growth to deplete the 2DEG layer [6],[7].

3. Hybrid MOSFET Structures:

Lateral GaN MOSFETs are being explored as an alternative because they offer normally-off operation and are less susceptible to reliability problems. By combining the high mobility of the 2DEG with MOS gate control, the GaN/GaN heterostructure in the RESURF region produces a hybrid MOS-HEMT that offers the potential for normally-off, low on-state resistance. [7].

III. CONVERTER TOPOLOGIES

A. DC/AC Converters

1. Single-Stage Buck–Boost:

A promising single-stage topology. By purposefully using a shoot-through state, which is normally prohibited in conventional VSI, the qZSI increases input voltage. By altering the duty cycle of the shoot-through, the quasi-Z-source network capacitors can be connected in parallel with energy storage (such as a battery) to control the state of charge. Multilevel structures, like the three-level NPC qZSI, can use the topology. [3],[8]

B. DC/DC Converters

These are important for voltage matching, boosting, and galvanic isolation, particularly in PV and automotive systems.

1. PV Module-Integrated Converters (MICs):

For galvanic isolation and voltage elevation in AC-module photovoltaic systems, a DC/DC stage with high frequency transformers is necessary. The design trend favors maximum use of parasitic elements and resonant power conversion to achieve high efficiency and power density. Examples provide advantages like high voltage gain and low input current stress. [5, 8].

2. Automotive DC/DC:

Bidirectional DC/DC converter are required in PHEVs that takes care of energy transfer between the high-energy battery

and the motor traction system. Fuel-cell vehicles use a DC/DC converter as a power conditioner [4],[5].

C. AC/DC Converters:

AC/DC conversion is typically handled by rectifier stages. For applications requiring regenerative capability and high power quality, Active Front End (AFE) rectifiers are used, often implemented with multilevel topologies. For instance, the downhill conveyor system uses 3 L-NPC back-to-back converters, enabling four-quadrant operation and regenerative braking due to bidirectional power flow [6][8].

D. AC/AC Converters (Matrix) :

This converter (MC) is a converter that uses bidirectional power switches and also is a direct AC/AC converter [6],[7].

1. Ideal Characteristics and Limitations:

Desirable features of MCs include generating unity power factor operation, and regeneration capability. Since the input is a voltage source, terminals must not be short circuited; conversely, since the load is typically inductive, output phases must never be opened[5],[7].

2. Implementation and Commutation:

The MC requires a fully controlled bidirectional switch. The main practical challenge is reliable current commutation due to the lack of natural freewheeling paths. This is addressed using intelligent multistep commutation strategies. The four-step semi-soft current commutation method relies on knowing the load current direction to ensure safe switching. This technique reduces switching losses by 50% due to soft switching in half the commutation process [5].

3. Modulation:

The standard direct transfer function method limits the maximum voltage ratio to 50%. By incorporating common-mode voltages (Venturini's optimum method), the ratio can be maximized. The indirect modulation method, based on a "fictitious dc link" concept, splits the conversion process into two conceptual stages (rectifier and inverter) to achieve a maximum ratio of 86.6% [6].

4. Practical Issues:

MCs require input filters (typically simple LC configurations) to reduce switching harmonics. Overvoltage protection typically employs a clamp circuit. The MC technology competes fiercely offering similar operating characteristics [7].

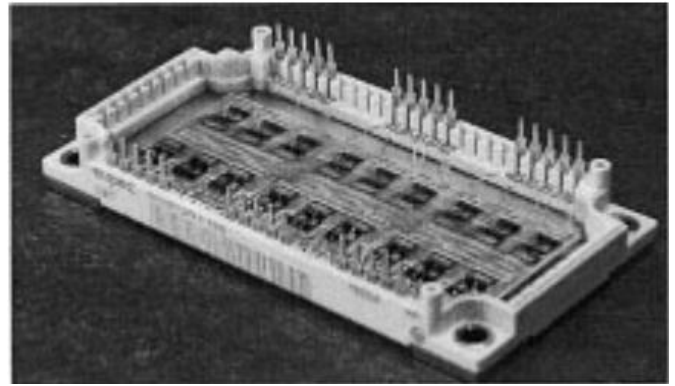


Fig. 2 The Eupec ECONOMAC matrix module [7].

IV. MULTI-LEVEL VOLTAGE SOURCE CONVERTERS AND MULTI-LEVEL CURRENT SOURCE CONVERTERS

A. Multilevel Voltage Source Converters (VSCs)

Classic VSC topologies include different varieties which are explained individually in the below sections [9].

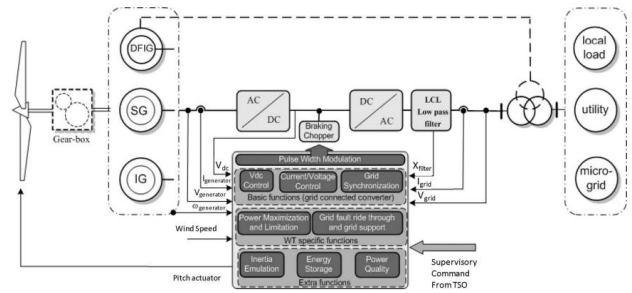
1. Classic Topologies and Constraints:

- *Neutral Point Clamped (NPC)*: The 3L(NPC) is widely commercialized due to its simple circuit structure. However, scaling NPC to higher levels leads to uneven loss distribution among inner and outer devices and difficulties with dc-link capacitor voltage balancing when using conventional modulation strategies with passive front ends [5],[9].
- *Cascaded H-Bridge (CHB)*: This topology is suitable for high-voltage operation using lower voltage switches and is also modular. The CHB typically requires a phase-shifting transformer to supply isolated DC sources, improving input power quality [6],[7].

2. Advanced VSC Topologies:

- *Active NPC (ANPC)*: The 3L-ANPC improves upon the 3L-NPC by replacing clamping diodes with controllable switches will give control over loss distribution, enabling substantially higher power rates (up to 200MVA commercially) [7],[8].
- *Modular Multilevel Converter (MMC)*: The MMC is modular and highly scalable for medium- and high-voltage applications. MMCs are commercially deployed in High-Voltage Direct Current transmission systems, reaching up to 1GW ratings. The MMC structure inherently solves the uneven voltage sharing problem associated with series-connected switches [8].
- *Transistor Clamped Converter (TCC) / Neutral Point Piloted (NPP)*: The NPP converter is the three-level TCC version, which uses a bidirectional switch for neutral point clamping, providing a controllable

eliminate the intermediate dc-conversion stage and improve global efficiency. WECSs are expected to contribute ancillary services, such as grid-frequency support (inertia emulation and droop control) [7],[8].



B. Photovoltaic (PV) Energy

1. Topologies:

The 3 L-NPC topology is advantageous for transformerless operation thus minimizing leakage currents. For large-scale PV plants entering the megawatt range, multilevel solutions like the 3 L-NPC and CHB-based interfaces are used to achieve higher voltages and better power quality. CHB is particularly appealing as the series connection of cells naturally elevates the voltage, potentially eliminating boost stages [4],[6].

C. Charging of EVs and HEVs

Power electronics must meet stringent requirements for efficiency, ruggedness, small size, and low cost in vehicles.

1. Architectures:

HEVs use two or more power/energy sources (e.g., engine and motor). PHEVs enhance HEVs because they add a high-energy battery pack and is capable of external charging [6],[9].

2. Conversion and Charging:

A battery charger is necessary for PHEV external charging. Because PEM fuel cells have a high-power density and can start up quickly. A power conditioner is used that matches stack voltage with the battery voltage [4],[7].

3. Device Requirements:

The automotive industry requires high-power devices, ideally SiC, that combine MOS gate control with low forward voltage drop $2V_f$ at 400A and high switching frequencies 10 kHz. The Power Control Unit (PCU) in systems like the Toyota Hybrid Synergy Drive II integrates multiple inverters and DC/DC converters into a compact assembly [5],[8].

B. Multilevel Current Source Converters (CSCs)

These topologies, such as the Load Commutated Inverter (LCI) and cycloconverter, historically dominated high-power, medium-voltage applications and it includes ship propulsion and grinding mills [9].

V. TOPOLOGY OF CONVERTERS FOR DIFFERENT APPLICATIONS

A. Wind Turbines (WECSS):

For multi- MW WECSs, back-to-back converters are the standard solution [10].

1. High-Power Conversion:

In the 1.53 MW range, the Doubly Fed Induction Generator (DFIG) using low-voltage IGBT back-to-back converters (rated at approx 30 of nominal power) is common. Above 5MW, handling high currents at low voltage becomes impractical [5],[8].

2. Medium-Voltage Multilevel:

Full-scale converters (FSCs) for direct-drive synchronous generators are increasingly implemented (Fig. 4 in source), often featuring IGCTs for medium-voltage operation 1kV 5kV. Commercial solutions, such as ABB's PCS6000 (up to 8 MW achieve efficiencies around 98%. A 6 MW five-level hybrid converter prototype based on ANPC converters has also been designed for high power density [6],[8].

3. Offshore Systems:

Offshore turbines if they are connected in series, leads to reduced transmission losses but requires conversion systems with extremely high efficiency and sophisticated control. Matrix converter technology is proposed for series arrays to

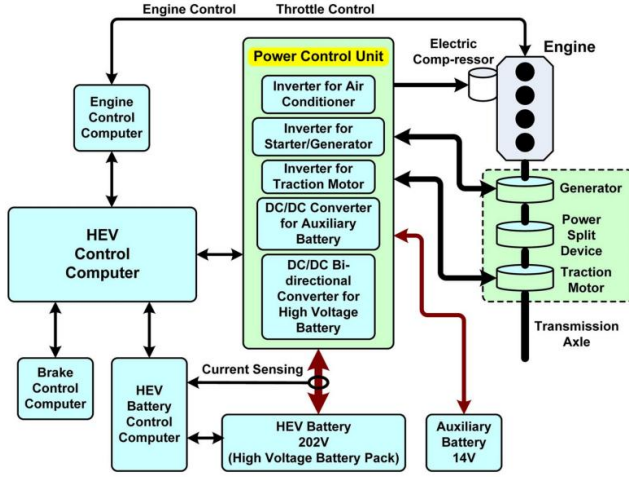


Fig. 5 Power control unit (Toyota Hybrid Synergy II) [10].

D. Ship Propulsion, Downhill Conveyor, etc.

1. Ship Propulsion:

While cycloconverters and LCIs dominate due to high power capability, multilevel NPC converters are replacing them in systems like large tanker propulsion due to superior power quality and performance[7],[10].

2. Downhill Conveyor:

Large downhill conveyor systems (e.g., 20MW) require four-quadrant regenerative operation, typically achieved using back-to-back 3L-NPC converters to maximize efficiency through bidirectional power flow [9],[10].

VI. POWER ELECTRONICS ALONG WITH PREDICTIVE CONTROL

Modulation techniques transform a reference voltage (u) into gate signals (S) to synthesize the desired output waveform. The switched waveform introduces undesirable harmonics, which must be kept below safety limits by balancing filter size and switching frequency [11].

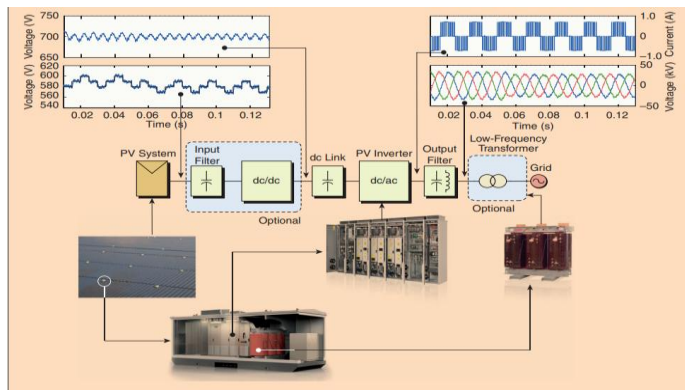


Fig. 6 Grid-connected PV system [11]

A. Traditional Modulation Techniques

1. Carrier-Based PWM:

This technique utilizes a carrier to compare the reference voltage which is a triangular carrier as well as uses high frequencies.

- *Phase-Shifted PWM (PS-PWM)*: Applied to modular converters (CHB, FC), it achieves to equalizes power losses, and distributes power evenly. This scheme is the only scheme of commercial modulation scheme utilized in CHB and FC is PS-PWM [8],[10].
- *Level-Shifted PWM (LS-PWM) and Phase-Disposition (PD-PWM)*: NPC topologies can use it. Because it eliminates harmonic components when line-to-line voltages are formed, LS-PWM is thought to be superior [8],[11].

2. Space-Vector Modulation (SVM):

SVM is conceptually equivalent to carrier-based PWM, yielding identical dc-link and harmonic performance. For multilevel converters, SVM exploits the redundancy of switching states (high degrees-of-freedom) to simplify computational cost and achieve auxiliary-control objectives, such as dc-voltage control or common-mode voltage minimization [9],[10].

3. Preprogrammed PWM (SHE/SHM):

- *Selective Harmonic Elimination (SHE)*: To remove particular low-order harmonics, switching angles are computed offline. The low bandwidth needed for control loops to follow the pattern limits SHE to low-dynamic power systems [6].
- *Selective Harmonic Mitigation (SHM)*: As a development of SHE, SHM incorporates a grid-code-based system of equations for the offline computation of switching angles. Compared to traditional SHE, this optimizes filter sizing and guarantees adherence to maximum harmonic limits [7].

B. Control Techniques with Implicit Modulator

Without the use of an explicit modulator, direct controllers are closed-loop control techniques that produce gate signals directly. The controller goal's error is minimized to determine the switching state [8].

C. Special Focus on Predictive Control

This is a powerful and simple method to control power converters [9],[11].

1. FS-MPC Mechanism:

By minimizing an objective cost function, FS-MPC finds the ideal switching state and forecasts the behavior of the VSI for each of the n possible switching states. The cost function can

include several goals, like monitoring phase currents, balancing dc voltages, or lowering switching losses [6].

2. Harmonic Performance and SVM:

Conventional FS-MPC produces a broad harmonic spectrum and a nonconstant switching frequency. However, FS-MPC is able to evaluate virtual states (linear combinations of finite states) due to the incredibly fast computational speed of contemporary microprocessors. This method, known as FS-MPC-SVM, produces a switching frequency that is closer to the sampling time while also improving the harmonic spectrum [8].

3. Predictive SHE/SHM:

This has been proposed to run SHE in a closed-loop, real-time operation. By using a predictive model that considers the real-time amplitude of each harmonic (via discrete Fourier transform), this method allows for higher dynamic performance capability than conventional offline SHE [10].

VII. RESULTS AND DISCUSSIONS

The state-of-the-art in power electronics is characterized by the accelerating deployment of advanced converter topologies enabled by Wide Bandgap (WBG) semiconductors, driven primarily by the search for increased efficiency. The physical limitations of conventional Silicon (Si) devices have been largely addressed, which can also operate at different conditions such as higher voltages, varying and higher temperatures, along with switching speeds [5],[6],[7].

Developmental results for SiC include the demonstration of 12 kV class PiN diodes and high-power SiC Gate Turn-off Thyristors (SiC-GTs) rated at 4.5kV, 120A operating in 100kVA inverters, while GaN research has achieved high-power GaN/GaN HFETs with breakdown voltages over 1.8kV on 4 inch Si substrates, as well as extremely high voltage HEMTs (8.3) on sapphire substrates [7],[8],[9].

Furthermore, the challenge of utilizing normally-on GaN HEMTs in power systems is being overcome through successful development of normally-off structures, such as those employing recessed-gate architectures or fluorine-based plasma treatment. In terms of topologies, the crucial practical implementation problem of commutation in AC/AC Matrix Converters (MCs) has been resolved by developing highly intelligent multistep commutation strategies facilitated by powerful modern digital devices, although the MC still struggles with ride-through capability. For DC/DC conversion, particularly in Photovoltaic (PV) systems, high-efficiency resonant converters are being integrated into Module Integrated Converters (MICs), with GaN devices being actively investigated for these high-power density applications [8],[9],[10].

Multilevel Converters (MLCs) remain essential for high-power utility interfaces; the commercial success of the 3L-NPC is due

to its simple power circuit structure, low component count, and minimal capacitors, although expanding DCC structures beyond three levels introduces significant challenges related to $\{ \text{uneven loss distribution} \}$ and difficulty in DC-link capacitor voltage balance using conventional modulation strategies with passive front ends. Application requirements dictate specialized topological choices: for multi-MW Wind Energy Conversion Systems (WECS) handling currents as high as approx 6300A (for a 7.5MW system at 690V, four-quadrant three-level back-to-back converters (often NPC's using IGBTs) are deemed appropriate to manage power density and high currents; meanwhile, large PV plants utilize multilevel topologies to enable MV grid connection and facilitate enhanced efficiency [10].

Finally, the evolution of modulation aims at greater performance across these complex topologies, with proving vital for modular MLCs (CHB, MMC) by providing necessary equal power distribution and loss equalization. Harmonically constrained grid operation benefits from Selective Harmonic Mitigation (SHM), which is an evolution of SHE focusing on meeting grid code limits rather than complete elimination, thereby reducing filter size and cost; this control philosophy can be efficiently implemented through a robust, powerful method capable of managing multiple objectives (like power flow and voltage balancing) by minimizing a cost function in real-time [3],[10].

VIII. CONCLUSION

Power electronics technology is rapidly maturing, driven by the shift toward high-power density, high efficiency, and stringent regulatory requirements for grid integration and electric transportation. The proliferation of specialized converter topologies (AC/AC Matrix, qZSI) and the industrial dominance of multilevel VSCs (NPC, CHB, MMC) reflect the increasing complexity and power demands of modern applications.

Future development will rely heavily on WBG devices (SiC and GaN) to exploit high switching speeds and higher operational temperatures, enabling reduced cooling requirements and size. Simultaneously, advanced control schemes, particularly Model Predictive Control, are essential for managing complex operational issues like voltage balancing and enforcing strict grid code compliance through techniques like Selective Harmonic Mitigation.

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